

TASK - 1

Writing Better Prompts

Demonstrating how prompt specificity improves AI-generated outputs

This task explores how refining a prompt — by specifying the audience, format, and level of detail — can dramatically improve the quality and usefulness of AI-generated responses. Three topics are examined, each with a vague prompt and an improved counterpart.

Topic 1 — Climate Change

● Vague Prompt

Prompt: Explain climate change

Output: Climate change is the long-term change in Earth's temperature and weather patterns. It is mainly caused by human activities such as burning fossil fuels, deforestation, and pollution. It can result in global warming, rising sea levels, and extreme weather conditions.

● Improved Prompt

Prompt: Explain climate change in 3 simple bullet points for a 12-year-old student.

Output:

- Climate change means the Earth is getting warmer over time.
- Pollution from cars, factories, and cutting trees causes climate change.
- It can lead to hotter weather, melting ice, and heavy rains.

Analysis: *The vague prompt produces a generic explanation without audience or format constraints. The improved prompt specifies the audience (12-year-old student), format (3 bullet points), and reading level, resulting in a clearer, more accessible answer.*

Topic 2 — Java Programming

● Vague Prompt

Prompt: Teach me Java

Output: Java is a programming language used for developing applications, websites, and software. It is platform-independent and widely used in software development. Java supports object-oriented programming and has features like inheritance, polymorphism, and encapsulation.

● Improved Prompt

Prompt: Teach me Java basics for beginners with simple examples in 5 easy points.

Output:

- Java is a programming language used to create applications and software.
- Java code is written inside classes and methods.
- Example: `System.out.println("Hello World");` prints text on the screen.
- Variables are used to store data like numbers and text.
- Java is beginner-friendly and used in many real-world applications.

Analysis: *The vague prompt delivers a high-level overview that assumes prior knowledge. The improved prompt targets beginners, requests structured numbered points, and asks for examples — resulting in a step-by-step, actionable introduction.*

Topic 3 — Fitness

● Vague Prompt

Prompt: Give fitness tips

Output: Fitness tips include regular exercise, healthy eating, drinking enough water, getting enough sleep, and maintaining a healthy lifestyle. Consistency is important for staying fit and healthy.

● Improved Prompt

Prompt: Give 5 beginner-friendly fitness tips for a college student who wants to stay healthy without going to the gym.

Output:

- Walk for at least 20–30 minutes daily.

- Drink enough water and avoid excessive junk food consumption.
- Perform simple home exercises such as push-ups or stretching.
- Aim for 7–8 hours of sleep each night.
- Stay active by taking stairs or walking more around campus.

Analysis: *The vague prompt returns broad, generic advice applicable to anyone. The improved prompt defines the target audience (college students), the quantity (5 tips), and a meaningful constraint (no gym required), yielding practical and relevant guidance.*

Comparison Summary

Topic	Vague Prompt	Improved Prompt	Key Improvement
Climate Change	Explain climate change	Explain in 3 bullet points for a 12-year-old	Audience + Format
Java Programming	Teach me Java	Basics for beginners with examples in 5 points	Level + Structure
Fitness	Give fitness tips	5 tips for college students without a gym	Context + Constraints

Key Takeaway: Effective prompt engineering depends on three core elements — defining the **audience**, specifying the **format**, and providing relevant **context or constraints**. Prompts that incorporate all three consistently produce outputs that are more focused, actionable, and professionally useful.